



India Road Accident Analysis: A Data Storytelling Approach Towards Safer Roads

Data Science & Visualization Project

A visual, evidence-driven study to identify patterns, prioritize interventions, and inform policy for safer roadways across India.



Problem Statement

Rising road safety crisis

India records a high and growing number of road accidents with significant human and economic cost.

Pattern discovery needed

Identifying temporal, geographic, vehicular, and environmental patterns is critical to reduce incidents.

Data-driven decisions

Evidence-based interventions outperform guesswork—analytics enable targeted policies and enforcement.

Role of visualization

Clear visual storytelling translates complex findings into actionable recommendations for stakeholders.

Approach – End-to-End Workflow



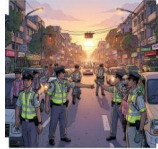
We followed a reproducible pipeline: collect reliable sources, clean and validate records, explore patterns, test hypotheses, create visuals, extract insights, and build a dashboard for stakeholders.

Scope – Who Benefits



Government Policy

Data to inform regulation, licensing, and infrastructure investment decisions.



Traffic Police

Targeted enforcement and resource allocation for high-risk zones and times.



Urban Planning

Design safer road geometry, pedestrian infrastructure, and intersection upgrades.



Public Awareness

Targeted campaigns for behavior change around peak-risk periods and driver types.

Objectives

1

Identify hotspots

Pinpoint geographic clusters and road segments with elevated accident frequency and severity.

2

Analyze trends

Detect seasonal and monthly patterns to reveal peak-risk windows.

3

Vehicle involvement

Examine which vehicle types and combinations contribute most to accidents.

4

Environmental factors

Assess influence of weather, light conditions, and road type on incident rates.

5

High-risk periods

Identify time-of-day and day-of-week risk windows for targeted intervention.

6

Actionable recommendations

Prioritize interventions with measurable impact—engineering, enforcement, and education.

Dataset Overview



Source: Kaggle — India Road Accident Dataset (link provided)

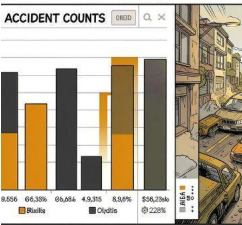
Description: Multi-year records of road accidents across Indian states with attributes covering location, date/time, vehicle types, road conditions, and severity.

Key attributes: State, District, Date, Time, Road_Type, Light_Conditions, Weather, Vehicle_Type, Victims, Severity, Latitude, Longitude

Sample rows (excerpt):

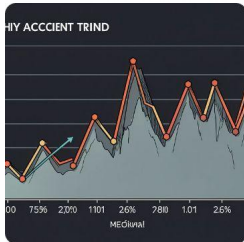
Date	State	Vehicle_Type	Severity
2020-07-14	Kerala	Two-wheeler	Fatal
2019-11-03	Maharashtra	Car	Injury
2021-01-22	Rajasthan	Truck	Damage only

Sample Visualizations – Mockups & Insights



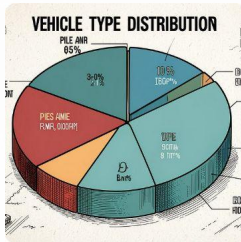
State-wise Accidents – Bar

Insight: A small set of states account for a disproportionately large share of accidents — focus interventions here maximizes impact.



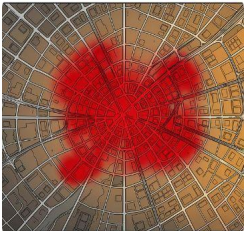
Monthly Trend – Line

Insight: Clear seasonal peaks during monsoon and festival months; timing enforcement and campaigns around these spikes is critical.



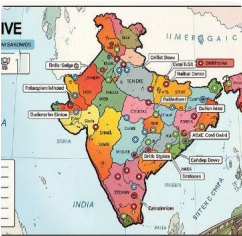
Vehicle Mix – Pie

Insight: Two-wheelers and cars form the majority of involved vehicles—safety measures should prioritize these user groups.



Hotspot Heatmap

Insight: Concentrated clusters around major intersections and arterial roads suggest engineering fixes and targeted enforcement.



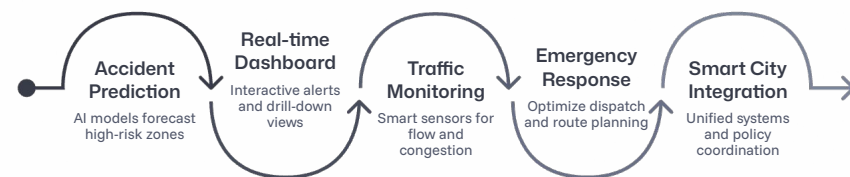
Interactive Map

Insight: Drill-down view enables local authorities to explore district-level trends and allocate resources precisely.

Expected Outcomes – KPI Cards

High-risk states List of top states by frequency and fatality rate to prioritize policy attention.	Peak accident months Identified months and festival periods with elevated incidents to time interventions.	Major causes Common causal categories: speeding, poor lighting, slippery roads, and mixed traffic conditions.
Accident trends Trends by vehicle type, road class, and environment to guide targeted engineering and enforcement.	Safety recommendations Actionable items: hotspot engineering, time-targeted enforcement, rider education, and emergency-response upgrades.	

Future Scope – Roadmap



AI-based prediction

Develop models to forecast high-risk times and locations using historical and live data streams.

Real-time dashboard

Deploy an interactive dashboard for authorities with alerts, filters, and drill-down analytics.

Smart monitoring

Integrate camera/IoT feeds and automated detection for violations and hazardous events.

Emergency optimization

Improve dispatch routing, hospital triage links, and first-responder coordination to reduce fatality rates.

Smart city integration

Embed findings into urban planning, mobility policy, and long-term infrastructure design.

Thank You

Dataset: Kaggle — India Road Accident Dataset

We welcome feedback from faculty and evaluators on methods, visualizations, and potential collaborations.



Thank you